



Improving aboveground biomass density mapping of arid and semi-arid vegetation by combining GEDI LiDAR, Sentinel-1/2 imagery and field data

Luis A. Hernández-Martínez^{a,b}, Juan Manuel Dupuy-Rada^{a,c}, Alfonso Medel-Narváez^d, Carlos Portillo-Quintero^e, José Luis Hernández-Stefanoni^{a,c,*}

^a Centro de Investigación Científica de Yucatán A.C., Unidad de Recursos Naturales, Calle 43 # 130. Colonia Chuburná de Hidalgo, C.P. 97200, Mérida, Yucatán, Mexico

^b Campo Experimental Todos Santos del Instituto Nacional de Investigaciones Forestales, Agrícolas y Pecuarias (INIFAP), Calle Agricultura No. 1555 Entre Calle México y Durango, La Paz, 23070, Baja California Sur, Mexico

^c Laboratorio Nacional CONAHCyT de Biología del Cambio Climático, SECIHTI, Mexico

^d Centro de Investigaciones Biológicas del Noroeste, La Paz, Baja California Sur, Mexico

^e Department of Natural Resources Management, College of Agricultural Sciences and Natural Resources Management, Texas Tech University, Lubbock, TX, 79401, USA

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ABSTRACT

Accurate estimates of forest aboveground biomass density (AGBD) are essential to guide mitigation strategies for climate change. NASA's Global Ecosystem Dynamics Investigation (GEDI) project delivers full-waveform LiDAR data and provides a unique opportunity to improve AGBD estimates. However, global GEDI estimates (GEDI-L4A) have some constraints, such as lack of full coverage of AGBD maps and scarcity of training data for some biomes, particularly in arid areas. Moreover, uncertainties remain about the type of GEDI footprint that best penetrates the canopy and yields accurate vegetation structure metrics. This study estimates forest biomass of arid and semi-arid zones in two stages. First, a model was fitted to predict AGBD by relating GEDI and field data from different vegetation types, including xeric shrubland. Second, different footprint qualities were evaluated, and their AGBD was related to images from Sentinel-1 and -2 satellites to produce a wall-to-wall map of AGBD. The model fitted with field data and GEDI showed adequate performance (%RMSE = 45.0) and produced more accurate estimates than GEDI-L4A (%RMSE = 84.6). The wall-to-wall mapping model also performed well (%RMSE = 37.0) and substantially reduced the underestimation of AGBD for arid zones. This study highlights the advantages of fitting new models for AGBD estimation from GEDI and local field data, whose combination with satellite imagery yielded accurate wall-to-wall AGBD estimates with a 10 m resolution. The results of this study contribute new perspectives to improve the accuracy of AGBD estimates in arid zones, whose role in climate change mitigation may be markedly underestimated.

1. Introduction

Forests can capture up to 60% of human carbon dioxide (CO₂) emissions (Pan et al., 2011). Therefore, international initiatives such as the Reducing Emissions from Deforestation and Forest Degradation (REDD+) strategy focus on protecting forests as a climate change mitigation measure. However, carbon capture and storage in forests is threatened by several phenomena, such as deforestation, extreme droughts, and increased frequency of fires (IPCC et al., 2019; Pan et al., 2011; Harris et al., 2021), which make forests act as an additional source of CO₂ emissions into the atmosphere (Friedlingstein et al., 2022).

To design and evaluate mitigation actions such as those implemented in REDD+, it is necessary to know the amount of biomass stored in forests and its spatial and temporal variation (Saatchi et al., 2011). However, estimating forest biomass is not straightforward. The amount of aboveground biomass (AGB) is not measured directly in the field but is estimated using allometric equations that relate tree diameter and height with the amount of AGB at the individual tree level. These mathematical models are used to estimate AGB at the plot or stand level by adding the biomass of the sampled trees and extrapolating the calculated biomass to larger areas. This process can lead to inaccurate estimates for various reasons, such as sampling representativeness,

* Corresponding author. Centro de Investigación Científica de Yucatán A.C., Unidad de Recursos Naturales, Calle 43 # 130. Colonia Chuburná de Hidalgo, C.P. 97200, Mérida, Yucatán. Mexico.

E-mail address: jl_stefanoni@cicy.mx (J.L. Hernández-Stefanoni).

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which is usually affected by forest heterogeneity (canopy cover, vertical structure, site quality, or management practices, among others), or limitations of access to remote sites. These limitations reduce the accuracy of AGB estimates when extrapolated at the landscape scale and hinder obtaining accurate national, regional, or global estimates (Kumar and Mutanga, 2019). Passive and active remote sensing has been used to make spatially explicit inferences of aboveground biomass density (AGBD) in forest ecosystems to obtain estimates at these scales (Theofanous et al., 2021; Li et al., 2020; Singh et al., 2023). However, the accuracy of an AGBD estimate is influenced by several factors, including the discrepancies between the spatial resolution of images and the size of plots; the time lag between the field data collection and image acquisition; the limited availability of plot data in certain regions or vegetation types, and the propagation of errors, among other factors (Réjou-Méchain et al., 2019; Ferrer-Velasco et al., 2022).

In Mexico, the information available in the National Forest Inventory (NFI) has made it possible to estimate AGBD with different remote sensors, reporting errors greater than 50.0% (Cartus et al., 2014; Rodríguez-Veiga et al., 2016; Urbazaev et al., 2018). A key aspect for increasing the quality of estimates lies in the need to increase the quantity and quality of the field data used to train the models. It has been pointed out that the NFI does not measure individuals with a DBH lower than 7.5 cm, which leads to underestimating AGBD, mainly in tropical dry forests and secondary forests, characterized by a higher density of small individuals (Rodríguez-Veiga et al., 2016; Hernández-Stefanoni et al., 2020). Similarly, NFI-based AGBD estimates could be biased in arid zones due to the underestimated contribution of shrubs with multiple stems that are branched below the DBH (1.3 m). Columnar cacti are also common or dominant in arid areas, but their contribution to AGBD storage may be underestimated due to their undersampling and lack of specific allometric equations. However, some studies indicate that cacti can harbor up to 11% of AGBD in arid zones (Búrquez et al., 2010).

LiDAR data penetrate the vegetation canopy and provide information on forest structure in remote areas not inventoried by the NFI; therefore, these data could help estimate AGBD more accurately. For this reason, LiDAR data have been proposed as an alternative to indirectly increase the number of samples available to train AGBD models with satellite imagery (Urbazaev et al., 2018; Coops et al., 2021; Singh et al., 2023). This is especially useful in arid zones of Mexico, which are underrepresented in the NFI. Although the high cost of LiDAR data collection remains restrictive, the availability of this data has increased in recent years thanks to NASA's Global Ecosystem Dynamics Investigation (GEDI) project. GEDI is a full-waveform LiDAR sensor designed to generate information about the structure of terrestrial ecosystems by sampling the vertical profile of the canopy. Since 2019, it has generated information in footprints of ~25 m in diameter (spatially discontinuous) located between $\pm 51^\circ$ latitude (Dubayah et al., 2020). The instrument has three laser sensors, one of which is divided into two beams (coverage beams), while the remaining two operate with full power (power beams; Dubayah et al., 2020). Potential applications include combining it with data from other optical and Synthetic Aperture Radar (SAR) sensors, with the aim of estimating AGBD with full coverage (wall-to-wall) over large areas (Shendryk, 2022; Liang et al., 2023; Tamiminia et al., 2024).

GEDI includes AGBD estimates for footprints in the GEDI-L4A Product (hereafter GEDI-L4A). To derive these estimates, global models were fitted relating field biomass with height metrics obtained from simulated GEDI footprints for different vegetation types (deciduous broad-leaved forests, evergreen broad-leaved forests, coniferous perennial forests, and a combination of grasslands/shrublands/wetlands) (Duncanson et al., 2022). However, these global models have certain weaknesses, such as the lack of representativeness of training data for some regions of the planet (Dubayah et al., 2022; Duncanson et al., 2022). For example, the global model integrated grasslands, shrublands, and wetlands into a single type of cover, for which only 65 training samples distributed in 5 countries were used, reporting an error

of 11.41 tons/ha (58%). This produces uncertainty about the use of GEDI-L4A AGBD estimates at regional or local scales. Therefore, the need to develop new local models that relate field data to GEDI height and coverage metrics has been proposed (Bullock et al., 2023; Dorado-Roda et al., 2021; Pascual et al., 2023). However, the georeferencing error of GEDI footprints, which is approximately 10 m (Dubayah et al., 2022; Tang et al., 2023), can lead to discrepancies between field-plot data and GEDI footprints.

GEDI-L4A can be used to construct wall-to-wall AGBD maps for different areas of the world, relating biomass estimates in footprints with variables derived from satellite imagery (e.g., Shendryk, 2022; Liang et al., 2023; Mohite et al., 2024). It has been suggested that the sensitivity parameter (associated with the penetration of footprints through the canopy) and the type of beam (coverage or full power) affect the quality of GEDI metrics and AGBD estimates at the footprint level (Kutchartt et al., 2022; Shendryk, 2022; Mohite et al., 2024). To reduce the effect of the footprint location error in relation to the pixels in a satellite image, different strategies have been proposed for grouping footprints in a given area (Shendryk, 2022). However, few studies have evaluated the development of new GEDI models based on field data, footprint selection and clustering, and wall-to-wall estimation with satellite imagery.

This study aims to provide a methodology to improve the accuracy of AGBD estimates for arid and semiarid zones in Mexico. Since the global models used to estimate AGBD in GEDI-L4A did not include training data for Mexico (Duncanson et al., 2022), the first objective of this study was to develop a model with field biomass data and height and cover metrics derived from GEDI to generate footprint-level AGBD (hereafter footprint AGBD) predictions for the state of Baja California Sur (BCS), Mexico. We expect that the model developed in this study will provide more accurate AGBD predictions than those derived from GEDI-L4A. The second objective was to evaluate models for wall-to-wall mapping of AGBD using three types of AGBD estimation as a dependent variable: 1) footprint GEDI estimates (grouped and filtered with different Sensitivity and beam-type parameters). This approach increases the sample size by estimating AGB values using a two-stage upscaling method: first, from field data to GEDI, and then from GEDI-derived AGB footprints to the study area using Sentinel-1/2 imagery; 2) estimates in NFI field plots, here we include an evaluation of the adjustment of NFI plots by incorporating the contribution of small plants to AGB and refining the biomass estimation of columnar cacti. These improvements aim to enhance alignment between field measurements and remotely sensed data, reduce AGBD underestimation, and increase the accuracy of satellite imagery-based estimates in arid zones; and 3) a combination of both sets of data, we anticipate that this combined approach will provide the most accurate AGB estimates, as it captures the variability of vegetation structure while effectively increasing the sample size. In these models, the metrics obtained from Sentinel-1 (S1) and Sentinel-2 (S2) satellite images were used as independent variables. Finally, the third objective was to map AGBD throughout the study area using the best model. Our expectation was that combining AGBD estimates at the GEDI footprint level and NFI field data related to satellite imagery would produce more accurate wall-to-wall estimates.

2. Materials and methods

2.1. Study area

The study was carried out in the state of Baja California Sur (BCS), located in northwest Mexico, between 22° and 28° N and -108° and -116° W (Fig. 1). Low-altitude areas are hot and dry, with mean annual temperatures ranging between 22°C and 24°C and mean annual precipitation of 200 mm. The mountain zones located in the south of the study area show ecological and environmental transitions related to the orography that gives rise to a more temperate and humid climate, with a mean annual precipitation of 700 mm in the highest areas (2090 m. a.s.

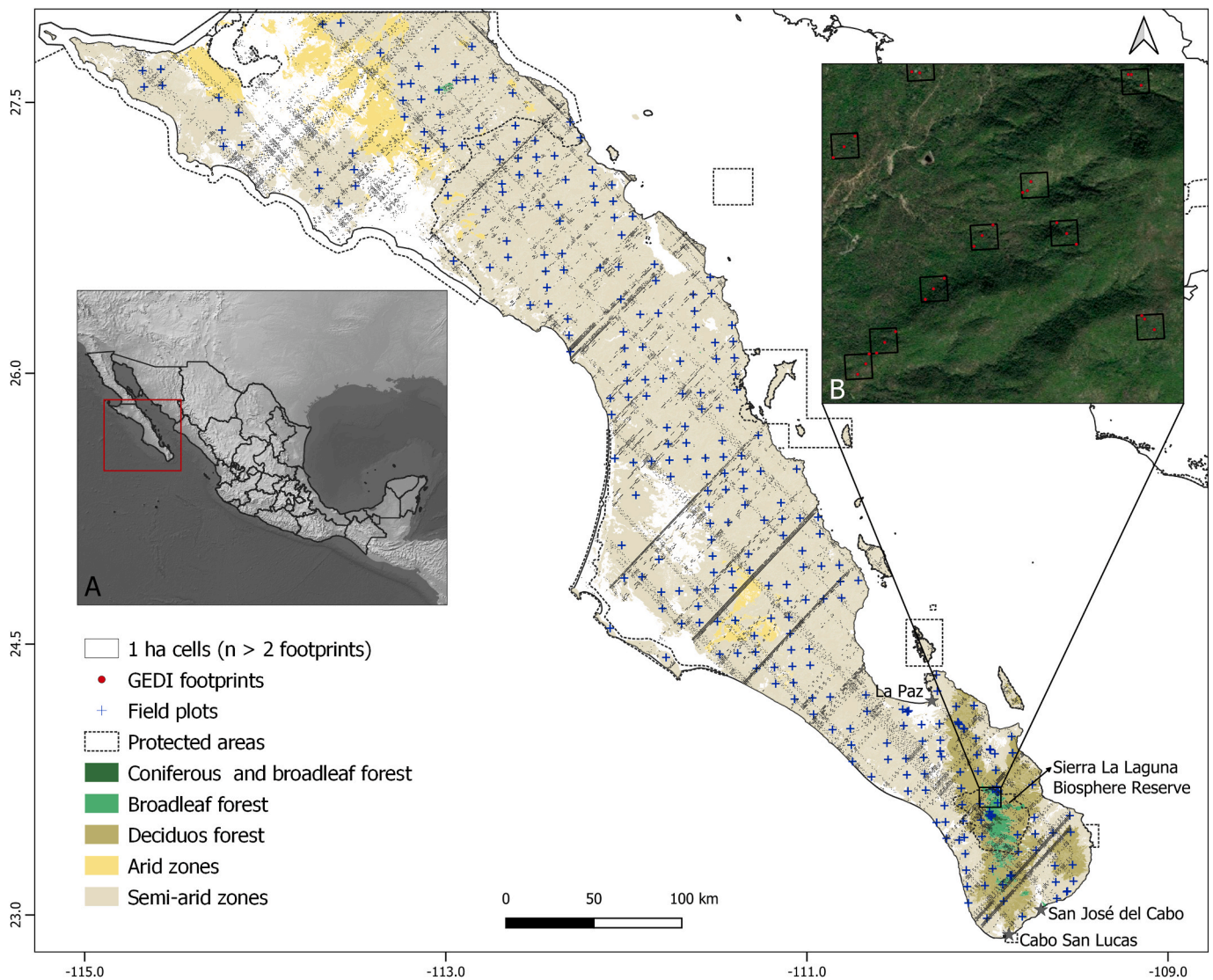


Fig. 1. Geographic location of the study area and the location of field plots (crosses) showing the main land cover types (different colours) and GEDI tracks (strait lines). A) Macro-location of the study area; B) Close-up of 1 ha cells with >2 GEDI footprints (red points).

l.; León-de la Luz et al., 2000; González-Abraham et al., 2010). Rains occur between July and January, driven by tropical cyclones from the Pacific between September and November. In BCS, plant associations such as xeric shrublands up to 500 m a.s.l., tropical deciduous forests between 500 and 1000 m a.s.l., and coniferous-broad-leaved forests above 1000 m a.s.l. Predominate (González-Abraham et al., 2010). The mountain range of Sierra La Laguna Biosphere Reserve, located to the south of the study area, in the Cape Region, plays a central role in biodiversity conservation and in the provision of a wide variety of environmental services, such as carbon storage and water harvesting. The main development centers are located in the south end of the state, in the cities of La Paz, San José del Cabo, and Cabo San Lucas. The growth of these cities, coupled with the accelerated development of tourism and productive activities such as extensive cattle ranching, are the main sources of impact on the terrestrial ecosystems of the state.

2.2. Field data

A field data survey was carried out in four vegetation types (xeric shrubland, tropical deciduous forest, broad-leaved forest, and a mix of coniferous with broad-leaved forest) during 2022. The sampling unit is a circular 1-ha area (radius = 56.42 m) composed of a cluster of four

circular plots with an area of 400 m² (radius = 11.28 m) with four concentric 80 m² subplots (radius = 5.04 m). The plots were distributed in an inverted "Y" shape, following the NFI sampling design (Comisión Nacional Forestal - CONAFOR, 2017). In each plot, all woody plants with a DBH (diameter of the stem at 1.3 m height) equal to or greater than 7.5 cm were measured. Additionally, in each subplot, we measured plants with a DBH lower than 7.5 cm and greater than 2.5 cm to incorporate smaller individuals in AGB estimates. The central coordinate of the 1 ha clusters was defined based on the location of a set of GEDI footprints corresponding to the post-rainy period (October to March) of 2020 and 2022. The footprints used met the quality parameters ("quality flags"; Dubayah et al., 2021a; Dubayah et al., 2021b).

For each individual, we recorded total height (m), stem height (cm), crown coverage (average of two perpendicular measurements, in m), basal diameter (cm), and diameter at breast height (DBH) (measured at 1.30 m, in cm). The survey was carried out after the rainy season, between September and December, comprising a sample size of 46 1-ha clusters.

Additionally, we used NFI data captured between 2015 and 2019 in BCS, including 280 1 ha clusters. The NFI field measurement methodology is similar to the one described above; however, unlike the plots measured in this study, the NFI does not include 80 m² subplots and does

not measure individuals with a DBH <7.5 cm, so the biomass contributed by these individuals is not considered. Furthermore, the NFI uses DBH as the only selection criterion, while this study used DBH and basal diameter.

2.2.1. Biomass calculation and correction of NFI data

AGB per individual was calculated in the clusters of plots measured in this study ($n = 46$) using allometric equations generated for arid zones. Additionally, we used equations to estimate AGB with additional dasometric variables (e.g., basal diameter and crown diameter; Table A1). Subsequently, we calculated the AGB per unit area (AGBD in Mg ha^{-1}) and the proportion of AGBD contributed by shrubs and small trees per AGBD percentile for each type of vegetation (Table A2). This proportion was used as a correction factor for the NFI data to account for the AGBD of small trees and shrubs that were omitted from the NFI sampling. Additionally, given the abundance of columnar cacti, such as *Pachycereus* sp and the lack of specific allometric equations to estimate their AGB, we calculated the individual AGB using the wood density reported by Romero et al. (2020).

2.3. Remotely sensed data

2.3.1. LiDAR data

Full-waveform LiDAR data from the GEDI sensor were used, which provide relative height (RHs; L2A) and canopy cover and vertical profile (L2B) metrics. RHs correspond to the height at which a certain percentile of energy is returned to the sensor relative to the terrain elevation (Dubayah et al., 2020). L2B metrics are Cover, defined as the percentage of the surface covered by the structural components of the canopy; Total Plant Area Index (PAI), represents one-half of the total structural elements of the canopy per unit area of the terrain; and Foliage Height Diversity (FHD_normal) is calculated using the PAI-normalized vertical structural components of the canopy (Tang and Armston, 2019). A set of footprints obtained from NASA's Land Processes Distributed Active Archive Center (LPDAAC) for the post-rainy season of 2021 and 2022 was used. These data were processed with the rGEDI package in R (Silva et al., 2020), extracting the quality parameters, the coordinates of each footprint, and the L2A and L2B metrics. The data were then filtered using the quality parameters ("quality_flag" and "l2b_quality_flag" = 1; "degrade_flag" = 0; Dubayah et al., 2021a; Dubayah et al., 2021b) and a "Sensitivity" threshold >0.95. The coordinates of each footprint were used to establish 46 one hectare plots distributed throughout the four main vegetation types studied (xeric shrubland, tropical deciduous forest, broad-leaved forest, and a mix of coniferous and broad-leaved forest).

Given that the GEDI footprints have a positioning error of approximately 10 m, a sufficiently large sampling unit (radius = 56.4 m) was employed during field work to minimize the impact of this error. Additionally, field plots were established in areas with homogeneous vegetation to ensure that each GEDI footprint accurately represented the corresponding vegetation type.

Subsequently, to increase the sample size thorough estimates of biomass from LiDAR data, we searched for GEDI footprints located within a one ha buffer matching NFI plots, considering the footprints for 2019 and 2020. This approach identified 7 additional plots with GEDI data. In total, a database was constructed with 53 field plots that matched the GEDI footprints.

2.3.2. Sentinel-1 images

C-band dual-polarization metrics of Sentinel-1 (VV and VH) SAR images with a 10 m spatial resolution were used. A mosaic of calibrated and ortho-rectified images was generated in the Google Earth Engine (GEE; Gorelick et al., 2017). Images were pre-processed in GEE, performing edge-noise corrections, speckle filtering, and radiometric terrain standardization according to the methodology by Mullissa et al. (2021). We used VV and VH polarizations, as well as the VV/VH ratio,

which is less sensitive to the effects of soil moisture and has been reported as a suitable AGBD predictor (Vreugdenhil et al., 2018; Shendryk, 2022).

2.3.3. Sentinel-2 images

This study used a mosaic of Sentinel-2 optical images derived from bands with 10 m spatial resolution (Blue, Green, Red, Near Infrared) with an L2A (below-atmosphere reflectance) processing level and low cloud cover (<10 %). The reflectance levels of these bands respond to the chemical and structural characteristics of the canopy, which are closely related to AGBD (Rodríguez-veiga et al., 2017). In addition to the bands, three Vegetation Indices (VI) were used: Normalized Difference Vegetation Index (NDVI), Enhanced Vegetation Index (EVI), and Soil-Adjusted Vegetation Index (SAVI). The three VIs are highly sensitive to the photosynthetic structures of vegetation. However, the SAVI offers the advantage of reducing soil effects in sites with low canopy cover (Huete, 1988), which is particularly relevant for arid zones. For both sentinel-1 and sentinel-2 imagery, 18 texture measurements were calculated in windows of 11×11 pixels, equivalent to $12,100 \text{ m}^2$, which is the area closest to the size of field clusters ($10,000 \text{ m}^2$). Texture metrics were calculated from the Gray Level Co-occurrence Matrix (GLCM; *glcmTexture* in GEE) around each pixel. Of the total texture metrics 14 metrics correspond to those proposed by Haralick et al. (1973), while four additional metrics were introduced by Conners et al. (1984). Texture metrics provide information about the variability of pixel values in a given area, so they can be used as indicators of structural variability of backscatter values in S1 images and of environmental variability of reflectance values in S2 images. Both structural variability and environmental variability are related to ecosystem productivity and, therefore, to AGBD (Fischer et al., 2019; Schweiger et al., 2018). The S1 and S2 mosaics were generated with images from the 2022 post-rainy season. In total, 39 S1 metrics and 133 S2 metrics were obtained (see Table A3).

2.4. Model fitting for AGBD estimation

2.4.1. Footprint AGBD estimation

In this study, AGBD models were developed in two stages (Fig. 2). In the first, a model was fitted using field clusters ($n = 53$) and GEDI metrics; this model was used to estimate footprint AGBD values (1,642,166 footprints). The objective of these AGBD estimates was to indirectly increase the sample size and then use GEDI footprints as field observations. The fitted model was compared with the footprint AGBD estimate reported by GEDI-L4A, which was predicted using the global AGBD models developed by Duncanson et al. (2022). This comparison used the AGBD values reported by GEDI-L4A on the same footprints as those used in the model fitted in this study.

The model was fitted with the Random Forest (RF) regression, which has been shown to be suitable for estimating AGBD, in addition to requiring a relatively simple parameterization (Tamiminia et al., 2024; Li et al., 2020). To avoid overfitting in the model, we calculated the correlation of AGBD with each explanatory variable and pre-selected only those variables with values of the Pearson correlation coefficient >0.4. Additionally, explanatory variables that were highly correlated with each other (Pearson correlation coefficient >0.9) were excluded. The importance of the selected variables was determined using two indicators: the percentage increase in the mean square error (%IncMSE) and the total decrease in the impurities of nodes from the division of the variables (IncNodePurity; Li et al., 2020). A total of 500 decision trees were used, and the optimal number of predictive variables for each node was defined using the *ModelMap* package in R (Freeman and Frescino, 2009; R Core Team, 2021). The model was validated through the cross-validation procedure (LOOCV; Picard and Cook, 1984). The accuracy of the model was evaluated with the coefficient of determination (R^2) derived from a simple linear regression between the observed and model-predicted values, the root mean square error (RMSE), the

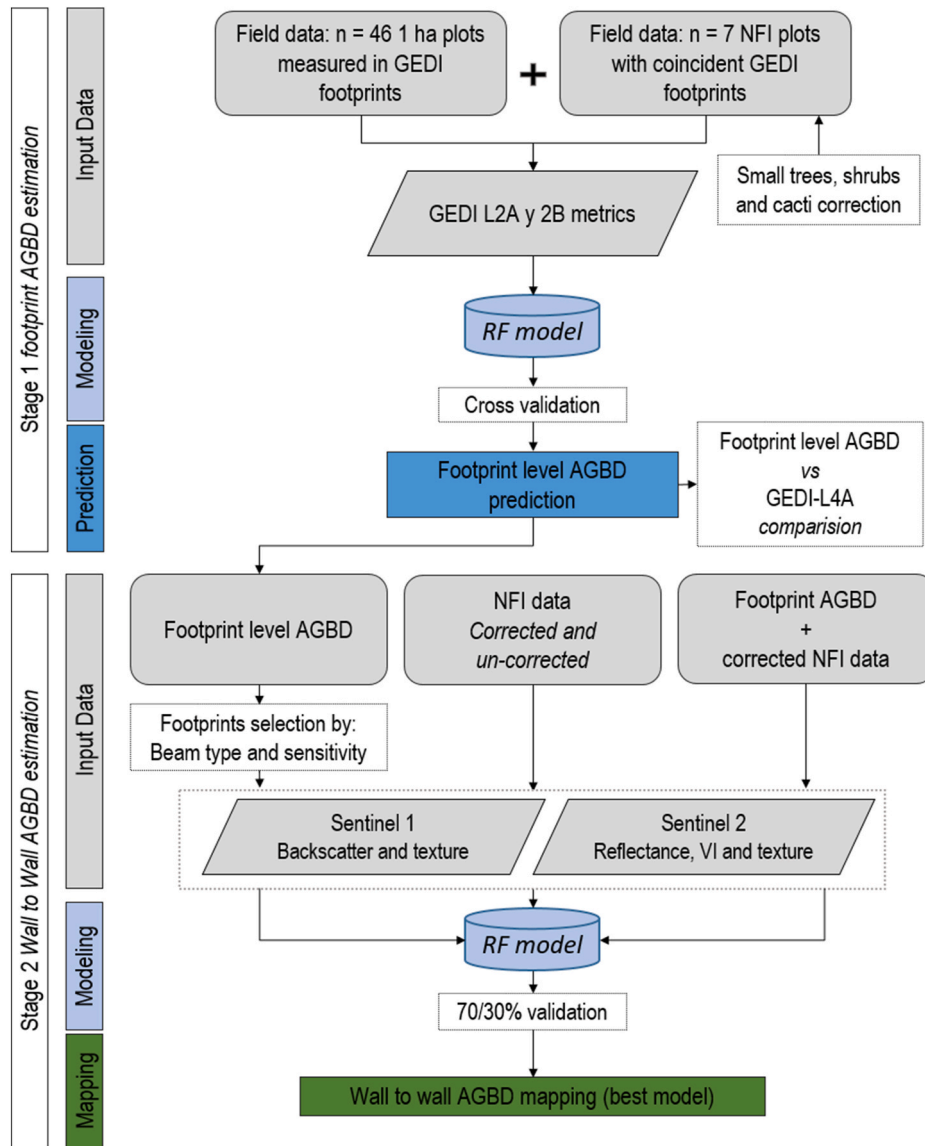


Fig. 2. Workflow for mapping AGBD: Stage 1 involves modeling at the footprint level using GEDI LiDAR data, while Stage 2 focuses on wall-to-wall mapping with Sentinel-1/2 imagery.

percentage RMSE (%RMSE), and the bias (BIAS). The latter yields values close to zero when the AGBD estimate is unbiased, negative values when the model tends to underestimate, and positive values when it tends to overestimate.

2.4.2. Wall-to-wall AGBD estimation

In the second stage of AGBD estimation the models that used three different types of AGBD estimates were compared. 1) AGBD estimated at the GEDI footprint level by the model fitted in the first stage. In this model, the best set of footprints for AGBD estimation was identified by constructing four sets of footprints considering the beam type (coverage or full power) and the Sensitivity parameter (>0.95 or >0.98 ; Table 1). Both parameters influence the AGBD estimation due to their relationship with the capacity of LiDAR pulses to penetrate the vegetation canopy. On the one hand, the power of the laser varies according to the beam type (cover or full power), while Sensitivity is directly related to the capacity of the pulse to reach the ground (Dubayah et al., 2020). In addition, the GEDI data location error makes it difficult to associate the position of each footprint with each pixel in the satellite imagery data (Shendryk, 2022; Mohite et al., 2024). To minimize this effect, footprints were grouped within 1 ha square cells; then, cells with >2

Table 1

Number of footprints and 1 ha cells with filtered AGBD estimates based on beam type and Sensitivity threshold.

Beam	Sensitivity	No. of footprints	No. of 1 ha cells with >2 GEDI footprints
Both	>0.95	133,316	39,560
Both	>0.98	2331	732
Full power	>0.95	131,280	38,949
Full power	>0.98	2307	726

footprints were selected, and the mean AGBD was estimated. 2) Models that used AGBD estimated using the NFI clusters. Here were considered both biomass corrected by small trees, shrubs and cacti and biomass without any correction. 3) In third model we combined both AGBD estimated with GEDI and estimated with corrected values of NFI data (Fig. 2). The three models used S1 and S2 metrics as predictive variables. Metrics were extracted by calculating the mean value of pixels located within 1 ha cells for GEDI estimation and using a 1 ha circular buffer for

NFI field data.

The second-stage models were also fitted with RF following a procedure similar to that used in the first stage; in this case, the models were validated by dividing the data into two subsets: training (70%) and validation (30%).

2.5. AGBD mapping and comparison with other studies

Using the most appropriate model of the second stage procedure, AGBD was mapped across the study area with the *ModelMap* package. Subsequently, using the validation dataset (30%), the average values of two previous studies that constructed AGBD maps for Mexico were extracted (Cartus et al., 2014; Rodríguez-Veiga et al., 2016). Both studies used environmental data (altitude), passive and active remote sensing (Landsat, MODIS, and ALOS-PALSAR), and NFI data recorded between 2004 and 2014. The accuracy of estimates was compared using R^2 , RMSE, %RMSE, and BIAS. Likewise, the mean of the estimates was calculated for each stratum defined according to the AGBD percentiles of the validation dataset, as well as the 95% confidence interval of the differences between the predicted and observed values.

3. Results

3.1. Footprint AGBD estimation

The model for estimating footprint AGBD outperformed the GEDI-L4A model, as evidenced by the validation results ($R^2 = 0.71$, RMSE = 26.93, %RMSE = 45.05, BIAS = 0.33 for our model compared to $R^2 = 0.48$, RMSE = 50.58, %RMSE = 84.60, BIAS = -34.97 for the GEDI-L4A model). The estimate derived from the global model reported in GEDI-L4A showed errors that almost doubled those of this study. In addition, the GEDI-L4A data showed a clear tendency to produce underestimated AGBD values (BIAS = -34.97; Fig. 3), while the model developed in this study reported a bias close to zero (BIAS = 0.33; Table 2; Fig. 3).

The best-fitted model included RH70, RH90, Cover, and PAI as explanatory variables. The most important variable was PAI, followed by Cover, RH70, and RH90 (Fig. 4). This model explained a high proportion of the variance in the data ($R^2 = 0.66$) (Table 2).

3.2. Models for wall-to-wall AGBD mapping

3.2.1. Models using AGBD estimated from GEDI data

The models developed from AGBD estimated by GEDI in 1 ha cells produced estimates with low RMSE values (below 17 Mg ha⁻¹; Fig. 5). The best goodness-of-fit parameters (in terms of R^2 and %RMSE)

Table 2

Model validation results to estimate footprint AGBD.

Model	Training		Test			
	R^2	RMSE	R^2	RMSE Mg ha ⁻¹	%RMSE	BIAS
Footprint AGBD model	0.66	28.86	0.71	26.93	45.05	0.33

corresponded to the models that only used footprints with Sensitivity >0.98 (both beams: $R^2 = 0.82$, %RMSE = 36.42; full-power beams: $R^2 = 0.84$, %RMSE = 34.47; Table 3; Fig. 5b and d, respectively). However, the lowest RMSE values were obtained with the footprint sets with Sensitivity >0.95 (Both beams: RMSE = 13.77 Mg ha⁻¹; full-power beams: RMSE = 12.91 Mg ha⁻¹; Fig. 5a and c, respectively; Table 3), which included a considerably larger number of samples (Table 1). In all cases, the BIAS was close to 0.

The Sensitivity threshold had an important effect on the sample size for predictions, decreasing the number of footprints by more than 98% when the 0.98 threshold was used (Table 1). The beam type had a marginal influence on the filtering of footprints, showing that in most cases, the coverage beams in the study area did not reach Sensitivity values higher than 0.95 or were excluded with the quality filters applied to footprints.

The model fitted only with full-power beams and a Sensitivity value > 0.95 was selected as the best model, considering the RMSE values, the sample size, and its representativeness in a wider range of AGBD values in the study area.

3.2.2. Models using AGBD estimated with NFI data

The models developed with corrected and uncorrected NFI AGBD performed poorly, accounting for only 6% and 10% of the variance with corrected and uncorrected AGBD, respectively (Table 3). Although the value of RMSE was higher for the corrected model (13.89 Mg ha⁻¹ compared to 8.48 Mg ha⁻¹ for the uncorrected model; Fig. 6), the % RMSE of both models was similar. An important difference between these models is that the AGBD correction for small trees, shrubs, and columnar cacti yielded higher and a broader range of AGBD estimates with an average predicted value of 16.57 Mg ha⁻¹, compared to 9.65 Mg ha⁻¹ for the uncorrected model.

A major disadvantage of the models fitted with NFI data is that field plots did not include areas with high biomass values (>50 Mg ha⁻¹). Therefore, they did not account for much of the AGBD variability in the study area, which may have significantly reduced the ability of these models to produce accurate predictions, even after correcting for small trees, shrubs and cacti.

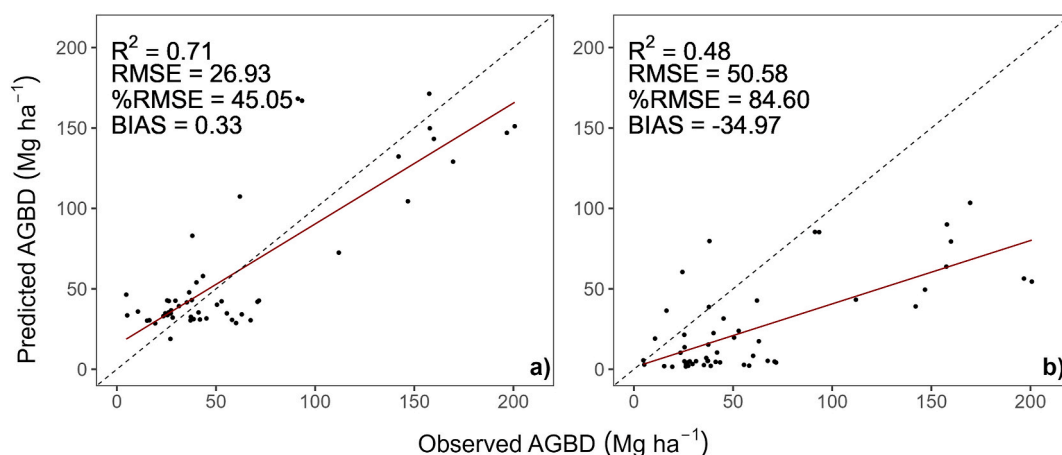


Fig. 3. Results of the validation of the fitted model to estimate footprint AGBD (a) vs the AGBD estimated by GEDI-L4A (b). The dotted line indicates the 1:1 ratio and the red line is the regression line fitted in the validation.

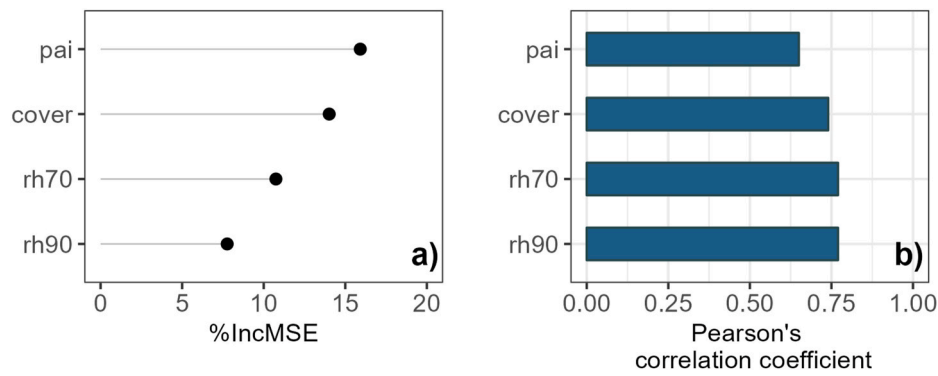


Fig. 4. a) Importance of AGBD model variables at the GEDI footprint level. b) Correlation of predictive variables with AGBD.

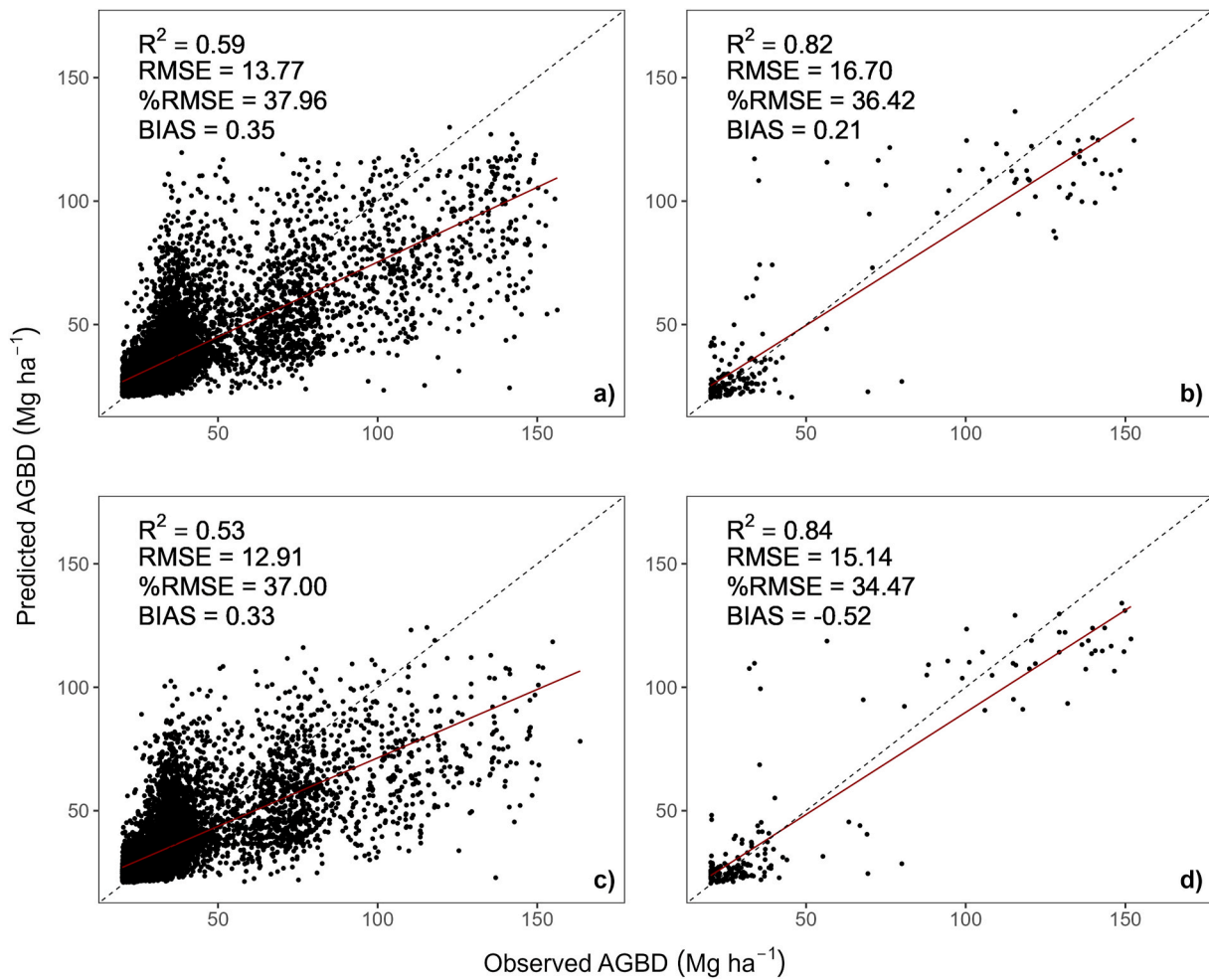


Fig. 5. Model validation for AGBD estimation from GEDI footprints and S1 and S2 images. a) Both beams and Sensitivity >0.95, b) Both beams and Sensitivity >0.98, c) full-power beams and Sensitivity >0.95, d) full-power beams and Sensitivity >0.98. The dotted line indicates the 1:1 ratio, and the red line is the regression line fitted during the validation.

3.2.3. Model combining AGBD estimates from GEDI and field data

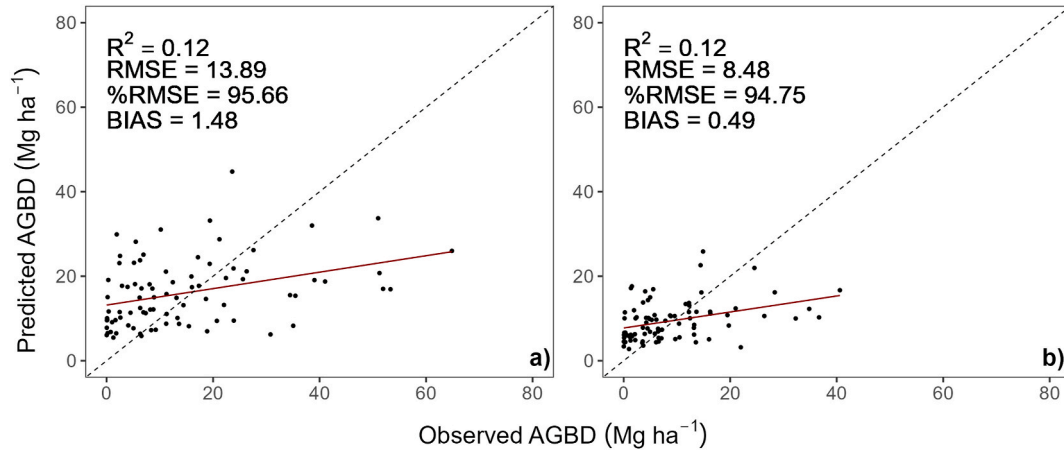
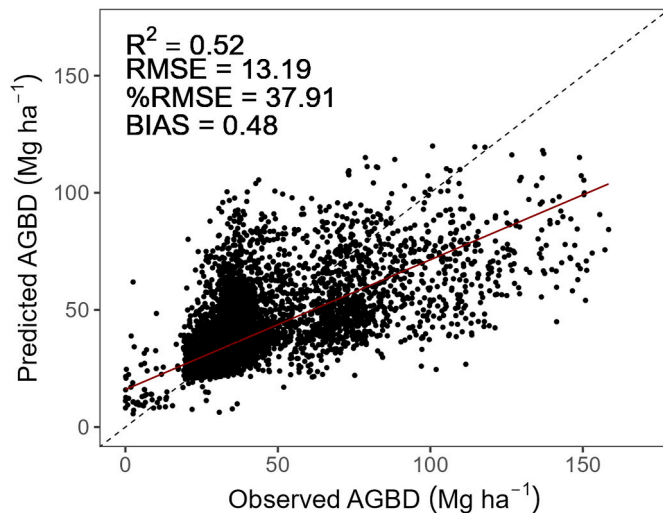
The combined model included AGBD estimates made in full-power beam footprints with Sensitivity >0.95 and corrected NFI data. This model explained a high proportion of the variance in AGBD ($R^2 = 0.53$) and performed well in the validation ($R^2 = 0.52$, RMSE = 13.19, % RMSE = 37.91; Fig. 7). The model parameters were similar to those of the best model obtained with the GEDI-estimated AGBD. Compared to the latter model, the combined model showed slightly larger values of RMSE and %RMSE. However, the combined model better represents the

variability of AGBD captured in the NFI, as it predicted a wider range of values (min = 5.6 Mg ha⁻¹, max = 120 Mg ha⁻¹) compared to the GEDI model (min = 20.89 Mg ha⁻¹, max = 124 Mg ha⁻¹). This broader range includes lower AGBD values corresponding to areas covered by xeric shrubland within the study area. This feature contributed to minimizing model overestimation in arid areas. Therefore, the combined model was chosen for AGBD mapping in the study area although the model calibrated exclusively with GEDI-estimated AGBD exhibited slightly better statistical performance.

Table 3

Model goodness-of-fit for AGBD estimation from GEDI footprints, field data, and S1 and S2 imagery.

Dependent variable	Subset	Training		Test		
		R ²	RMSE Mg ha ⁻¹	R ²	RMSE Mg ha ⁻¹	%RMSE
GEDJ-estimated AGBD	Both beams and Sensitivity >0.95	0.58	13.66	0.59	13.77	37.96
	Both beams and Sensitivity >0.98	0.86	15.10	0.82	16.70	36.42
	Full-power beams and Sensitivity >0.95	0.54	13.11	0.53	12.91	37.00
	Full power beams and Sensitivity >0.98	0.85	15.62	0.84	15.14	34.47
Field-measured AGBD using NFI	Corrected for small trees and cacti	0.06	17.66	0.12	13.89	95.66
	Uncorrected	0.10	9.75	0.12	8.48	94.75
	Full-power beams and Sensitivity >0.95 + corrected NFI	0.53	13.37	0.52	13.19	37.91

**Fig. 6.** Model validation for AGBD estimation from a) corrected and b) uncorrected NFI field data and S1 and S2 images. The dotted line indicates the 1:1 ratio, and the red line is the regression line fitted during the validation.**Fig. 7.** Validation of the combined field data model and GEDI AGBD estimation. The dotted line indicates the 1:1 ratio, and the red line is the regression line adjusted during validation.

The most important variable in the combined model was NDVI_savg (sum of NDVI averages), followed by B4_savg. Both variables were highly correlated with AGBD, positively in the case of NDVI_savg (0.48) and negatively in B4_savg (−0.48). However, SAVI_diss, which showed the highest correlation (0.54), was not particularly important in the model (Fig. 8).

In the map obtained from *ModelMap* using the best model (Fig. 9), the southern zone of the study area, specifically the Cape region mountain range and the Sierra la Laguna Biosphere Reserve (REBISLA),

showed the highest AGBD levels. Tropical deciduous forest, broad-leaved forest, and coniferous-broad-leaved forest predominate in this area. REBISLA reached an AGBD of 134.74 Mg ha⁻¹ in its highest zone (mean = 65.15 Mg ha⁻¹; Fig. 9a). In the Cape region, north of the mountain range where REBISLA is located, the Sierra de las Cacachilas also showed relatively high AGBD values (mean = 63.92 Mg ha⁻¹; Fig. 9b). Scattered patches of tropical deciduous forest are also located in this area. The Sierra de la Giganta, a region in the north of the study area, also showed relatively high AGBD values (mean = 55.08 Mg ha⁻¹; Fig. 9c) compared to low-altitude areas. As expected, low-altitude arid and semi-arid zones had the lowest AGBD levels. Regarding vegetation types, the mix of coniferous with broad-leaved forest yielded the highest AGBD values (mean = 89.26 Mg ha⁻¹), followed by the broad-leaved forest (mean = 72.59 Mg ha⁻¹), the tropical deciduous forest (mean = 49.08 Mg ha⁻¹), semi-arid xeric shrubland (mean = 38.19 Mg ha⁻¹), and arid xeric shrubland (mean = 24.44 Mg ha⁻¹) (Table 4). In terms of total AGB stored by vegetation type, arid xeric shrubland had the highest values with 218.59 million tons (M Ton), equivalent to 102.74 M Ton of C, followed by the tropical deciduous forest with 22.68 M Ton of AGB, equivalent to 10.66 M Ton of C. The contrast between the values of mean versus total AGBD and the rankings of the vegetation types is due to the marked differences in the area covered by each vegetation type in the study area (Table 4).

3.3. Comparison with other studies

The comparison of the AGBD estimates in this study with previously published AGBD maps clearly showed that the estimates by Cartus et al. (2014) and Rodríguez-Veiga et al. (2016) greatly underestimated the AGBD in the study area (Fig. 10a). The distribution of errors in AGBD percentiles showed that the combined model tended to produce slight overestimates in areas with the lowest AGBD (up to 37 Mg ha⁻¹), while the opposite trend is observed in sites with the highest AGBD. The

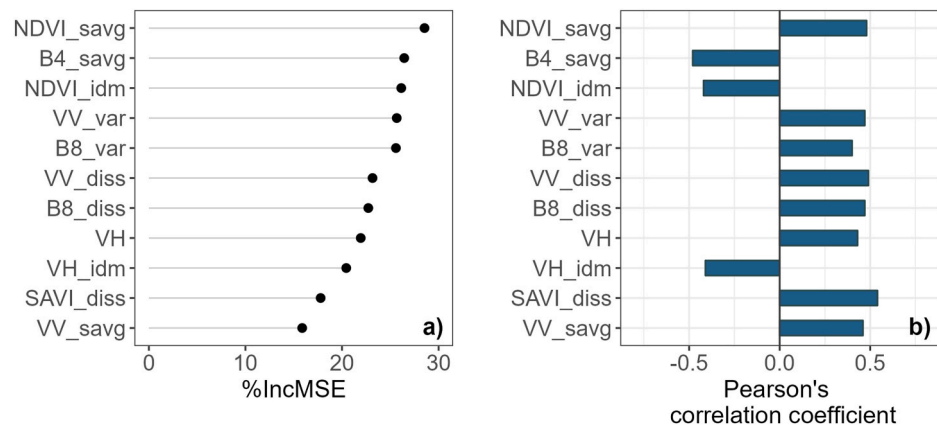


Fig. 8. a) Importance of the variables of the combined model. b) Correlation of the predictive variables with AGBD.

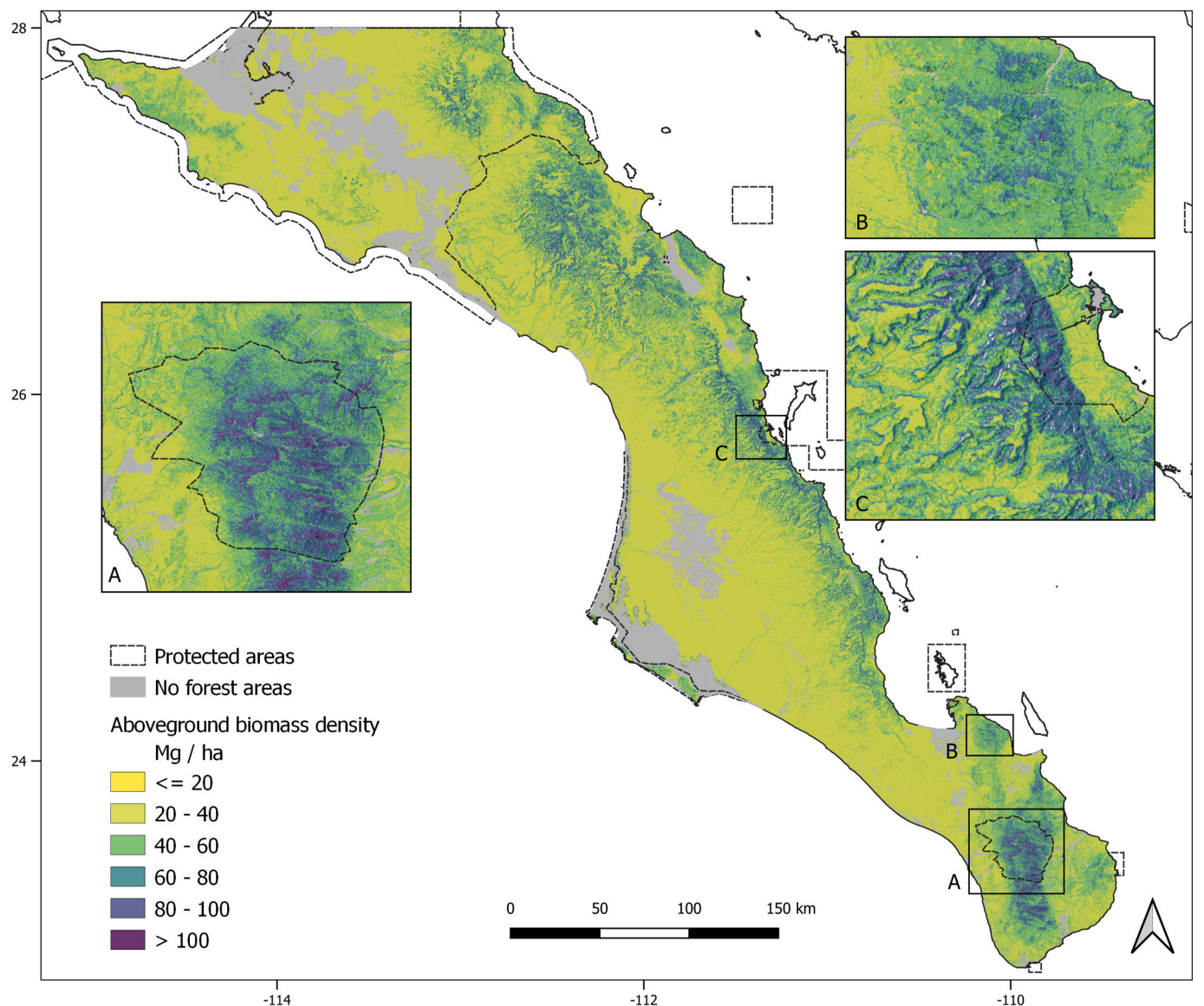


Fig. 9. Map of estimated AGBD from S1 and S2 images using the best model obtained. A) Sierra la Laguna Biosphere Reserve; B) Sierra de las Cacachilas; C) Sierra de la Giganta.

Table 4
Aboveground and C biomass density by vegetation type in BCS.

Vegetation type	Area (ha)	Mean AGBD (Mg ha ⁻¹)	Total AGB (M ton)	Total C (M ton)
Coniferous-broad-leaved forest	4368.79	89.26	0.38	0.18
Broad-leaved forest	49,978.95	72.59	4.05	1.90
Deciduous forest	396,242.89	49.08	22.68	10.66
Arid zones	404,530.85	24.44	11.08	5.21
Semiarid zones	5,124,369.00	38.19	218.59	102.74
Other vegetation	714,963.24	40.38	20.30	9.54
Total ^a	6,694,453.72	52.32	277.07	130.22

M ton: million tons.

^a The total area does not include the vegetation types that were excluded from the estimate, such as urban and agricultural areas, water bodies, and areas with bare soil.

studies by [Cartus et al. \(2014\)](#) and [Rodríguez-Veiga et al. \(2016\)](#) produced considerable underestimates throughout the entire AGBD range, which increased gradually as the AGBD increased, reaching underestimation levels above 50 Mg ha⁻¹ ([Fig. 10b](#)). The errors estimated with the validation dataset (RMSE = 35.58 Mg ha⁻¹ and 38.12 Mg ha⁻¹ for [Cartus et al., 2014](#); [Rodríguez-Veiga et al., 2016](#), respectively) almost tripled the error of the combined model (RMSE = 13.19 Mg ha⁻¹). The same trend was observed in %RMSE ([Fig. 10c](#)). The dispersion of the predicted values based on the observed and BIAS values also clearly showed the underestimation of the AGBD maps that preceded this study

([Fig. 10c](#)).

4. Discussion

This study seeks to provide a methodological approach that improves the accuracy of AGBD estimation in arid and semi-arid areas of Mexico. The results obtained clearly show that the proposed methodology improved the accuracy of previous estimates that considerably underestimated AGBD in Baja California Sur. This allows us to reassess the importance of arid and semiarid vegetation in mitigating climate change and design better-informed REDD + strategies. The proposed approach consists of two stages and several components, the results of which are discussed in detail below.

The use of GEDI data in arid zones has been little explored, despite the limitations of global GEDI-L4A models in predicting aboveground biomass, as they were calibrated with very few samples ([Shendryk, 2022](#); [Duncanson et al., 2022](#)). Until now, no study had evaluated the accuracy of GEDI-L4A estimates in arid zones using field data. This study demonstrated that calibrating local models by linking field data with GEDI and incorporating additional variables beyond RHs as predictors of AGBD helps reduce biases in estimates for arid zones, decreasing estimation errors compared to AGBD predictions in the GEDI-L4A product.

The estimation of AGBD in arid zones using satellite imagery is particularly challenging due to various factors, such as the presence of an open canopy and the limited availability of field data ([Eisfelder et al., 2012](#)). The NFI of Mexico allocates smaller sample sizes and more

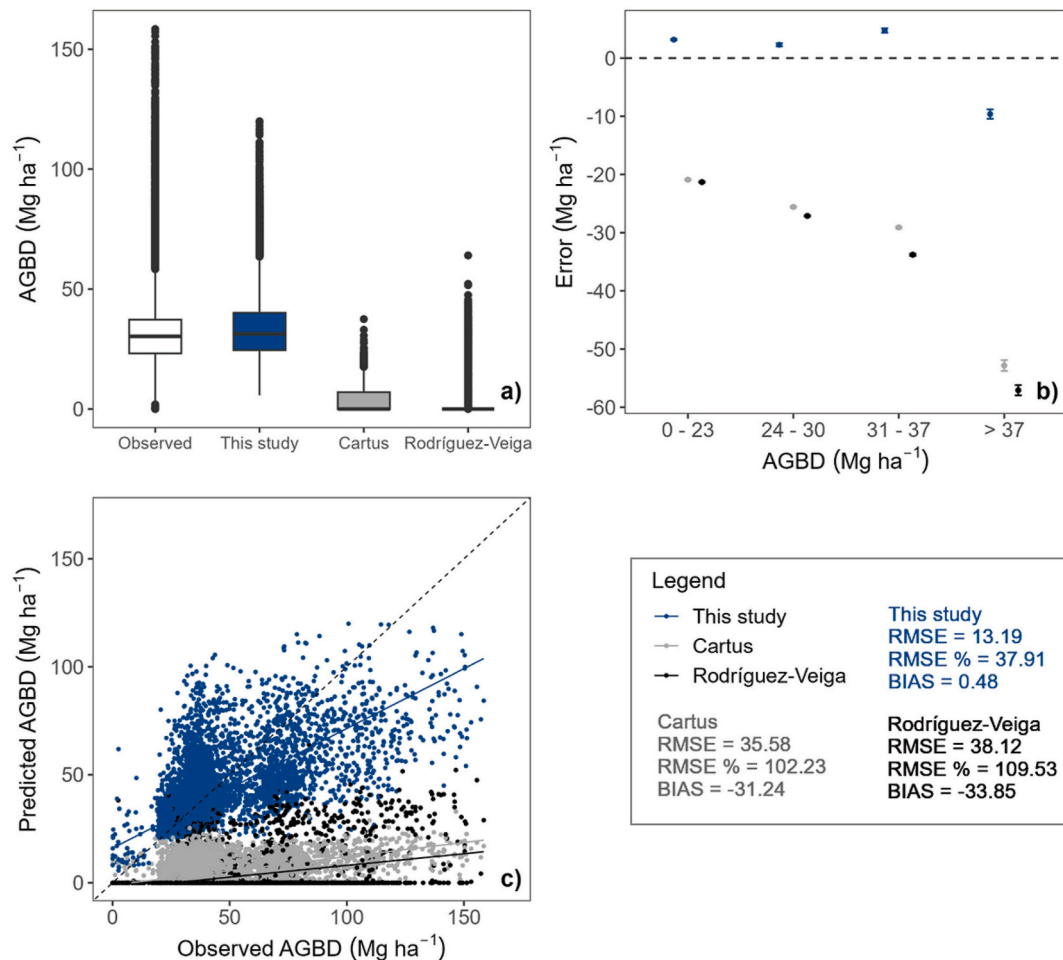


Fig. 10. Comparison of AGBD reference values in the study area (Observed) with estimates in this study (blue) and the maps of [Cartus et al. \(2014\)](#); gray) and [Rodríguez-Veiga et al. \(2016\)](#); black). a) General distribution of estimates; b) mean error with respect to corrected NFI data and 95% confidence interval; and c) dispersion of observed values relative to predicted values, the RMSE, %RMSE, and BIAS. Solid lines are regression lines; the dotted line represents a 1:1 ratio.

widely spaced plots in arid zones (Comisión Nacional Forestal - CONAFOR, 2017), compromising the representativeness of the sample. This study presents a methodological approach that indirectly increased the sample size by estimating AGBD using GEDI data calibrated with field data, in addition to correcting AGBD values in NFI plots for not measuring small trees and refining biomass estimations of columnar cacti. The model that relates the combination of AGBD estimates from GEDI and corrected NFI field with Sentinel-1/2 imagery data provided the most representative and accurate AGB estimates by capturing the variability in vegetation structure and effectively increasing the sample size. This approach improved the accuracy and significantly reduced the underestimation of AGBD in arid zones. Furthermore, the availability and performance of footprints with different quality parameters were also evaluated to optimize biomass estimation with GEDI in vegetation with an open canopy, an aspect that has been little studied.

4.1. Footprint AGBD estimation

The accuracy of the model developed in this study to predict footprint AGBD was considerably higher than that of GEDI-L4A (Fig. 3). In the latter model, AGBD was estimated with the global model of grasslands, shrublands, and wetlands ($R^2 = 0.86$, %RMSE = 55.39; Duncanson et al., 2022) and continental deciduous broad-leaved forests ($R^2 = 0.66$, %RMSE = 38.08; Duncanson et al., 2022). The error of the model obtained in this study (%RMSE = 45.05) was greater than the deciduous broad-leaved forest model and lower than the grassland, shrubland, and wetland model. However, compared to our model, GEDI-L4A estimates involved a greater underestimation, especially in clusters of plots with the highest AGBD (Fig. 3), which correspond to broad-leaved and the mix of coniferous with broad-leaved forests. These results support our first prediction that the model generated to estimate footprint AGBD in this study would be more accurate than the GEDI-L4A model. This is largely because the continental models did not include training data located in Mexico (Duncanson et al., 2022). In particular, the grassland, shrubland, and wetland model was fitted with only 65 samples from other regions of the world (Australia, Spain, New Zealand, Tanzania, and the USA; Duncanson et al., 2022) but did not include arid and semi-arid vegetation of Mexico.

Furthermore, the use of L2B canopy coverage and vertical profile metrics (Cover and PAI), which were not used in the GEDI-L4A models (Duncanson et al., 2022), could have contributed to improving the performance of the fitted model in this study. Other studies reported improvements in model performance to predict footprint biomass when GEDI L2B variables were included (Dorado-Roda et al., 2021). However, the use of Cover and PAI metrics in arid zones had not been reported until now. Given the strong influence of vegetation structure on AGBD prediction using full-waveform LiDAR data (Bruening et al., 2021), including such variables may be key to improving the accuracy of the models (Jia et al., 2024). Recent studies validating GEDI-L4A predictions have reported its tendency to underestimate AGBD (e.g., Indirabai and Nilsson, 2024; Li et al., 2024; Jia et al., 2024), which highlights the need to develop new models such as the one presented here.

The sample size to fit the models in this study was relatively small ($n = 53$) and did not allow us to fit models by vegetation type. Although other studies have developed models that integrate different vegetation cover values (e.g. Bullock et al., 2023; Duncanson et al., 2022), the accuracy of GEDI AGBD predictions has been shown to vary considerably between forest types (Jia et al., 2024). In this study, errors close to those of other stratified models were obtained, such as those of Duncanson et al. (2022) and Dorado-Roda et al. (2021) (%RMSE: 28.66–78.94 and 38.17–84.74, respectively). In addition, the geolocation errors of GEDI footprints strongly influence the accuracy of biomass estimates (Milenković et al., 2017; Roy et al., 2021; Tang et al., 2023). To mitigate the geolocation error, this study obtained field samples that match the specific location of GEDI footprints by using a plot large enough to

ensure that the footprint is covered by covers the area of the field plots.

Additionally, the temporal discrepancy between field data collection and GEDI footprints was minimized in this study using footprints from the same year or the preceding year during the same phenological season. However, to increase the sample size, the period for the search for footprints matching NFI field plots should be extended. The temporality and location criteria used in this study allowed us to obtain only 7 GEDI footprints that match NFI data. Although reducing these constraints can substantially increase the sample size, the temporal discrepancy could lead to increased underestimation of AGBD (Rodríguez-Veiga et al., 2016; Bullock et al., 2023). Future studies should assess and compare the magnitude of these two sources of error.

4.2. Models for wall-to-wall AGBD mapping

4.2.1. Models developed with GEDI data

In addition to the footprint estimate (GEDI-L4A), the AGBD reported by GEDI includes an estimate in 1 km^2 cells (GEDI-L4B), which is carried out through a hybrid inference considering the estimation error from the distribution of sensor footprints (Patterson et al., 2019); however, it offers a map with discontinuous coverage. In contrast, this study integrated footprints in 1 ha cells, which were subsequently associated with variables derived from optical and SAR imagery. This procedure has several advantages; first, it reduces the geolocation errors of GEDI footprints by estimating the average AGBD in 1 ha, as shown by other studies (Hernández-Stefanoni et al., 2018; Frazer et al., 2010); second, it produces an AGBD map with a 10 m spatial resolution and third, it provides continuous wall-to-wall coverage.

Other studies have used GEDI footprints combined with satellite imagery to map AGBD through a similar approach to that in the present work (Shendryk, 2022; Liang et al., 2023; Mohite et al., 2024). However, in contrast to this study, these models were fitted using GEDI-L4A AGBD estimates as a dependent variable, and none of them generated a new model for estimating footprint AGBD. Shendryk (2022) fitted models using GEDI-L4A in combination with environmental data and S1 and S2 imagery for the USA and Australia, reporting %RMSE values of 41 and 77, respectively. On the other hand, Liang et al. (2023) mapped AGBD in a dry forest of Mozambique using GEDI-L4A and Landsat imagery, obtaining an %RMSE of 42.31. These two studies grouped different types of vegetation in the models and included areas of shrubland with low AGBD values similar to those in the study area (mainly in the Australian model; Shendryk, 2022). The error obtained with the model developed in the present study (%RMSE = 37.91) was lower than the error reported in the above studies; this improvement could be attributed to the use of a new model to predict the footprint AGBD in the study area.

This study found that increasing the Sensitivity threshold of GEDI-L4A footprints improved the fit of the models (Table 3), similar to the findings of Liang et al. (2023) and Mohite et al. (2024). This could be because a higher Sensitivity is associated with a greater ability of footprints to reach the ground and thus generate better quality metrics (Dubayah et al., 2020). However, since this procedure significantly decreased the sample size and its spatial distribution in the study area, and considering that a Sensitivity threshold above 0.90 may be sufficient for GEDI footprints to cross the forest canopy (Hofton et al., 2019), this study used the model with a Sensitivity threshold above 0.95. On the other hand, in contrast to our observations with the Sensitivity threshold, the beam type had little influence on the sample size (Table 1), suggesting that the coverage beams generated few footprints that exceeded the quality filters (quality flags or Sensitivity) in the study area. This could be because shrublands and some areas of tropical deciduous forest exhibit open canopies, which, due to the effects of soil, makes it difficult to obtain good-quality footprints in the coverage beams (Dorado-Roda et al., 2021; Duncanson et al., 2022).

4.2.2. Model constructed with NFI data

It has been shown that the exclusion of small trees induces

underestimation errors in AGBD models (Rodríguez-Veiga et al., 2016; Hernández-Stefanoni et al., 2020). For this reason, the present study evaluated the effect of including a correction factor for small trees, shrubs, and cacti in the NFI data. Nonetheless, the fitted models with and without correction performed poorly (%RMSE >94).

Contrary to the second prediction of this study, the corrections made to the NFI data did not improve the performance of the models. This could be attributed to some drawbacks of NFI data in the study area. First, the sampling design limits the number of plots for drylands (plots arranged in a 10 km × 10 km grid), compared to tropical and coniferous forests (5 km × 5 km; Comisión Nacional Forestal - CONAFOR, 2017). Second, the small total area and limited accessibility of broad-leaved and mixed broad-leaved and coniferous forests in BCS limited the NFI sampling in zones with high AGBD values within the study area. The NFI data used in this study included only four plots in broad-leaved and the mix of coniferous with broad-leaved forests, which is equivalent to approximately 1% of the sample. As a result, the sample was not representative of the entire study area. To have more representative and accurate AGB field estimates, it is necessary to increase the sample size (e.g., by reducing the size of NFI grids).

4.2.3. Combined model

The combined model performed similarly to the one constructed only with GEDI AGBD estimates. However, the latter tends to overestimate aboveground biomass density in areas with low AGBD. These results are consistent with those of Shendryk (2022), who reported a tendency to overestimate AGBD in shrubland areas with low AGBD values. In the present study, the use of the combined model that integrated corrected field data from the NFI and GEDI AGBD estimates in 1 ha cells made it possible to substantially reduce the degree of overestimation in xeric shrubland areas, which make up over 75% of the study area.

In the combined model, the variables of both types of sensors (S1 and S2) were good AGBD predictors, with NDVI_savg, B4_savg and NDVI_idm as the most important, followed by VV_var. In most cases, the best predictors were texture variables, which were both positively and negatively correlated with AGBD (Fig. 8), as reported in other studies that included different vegetation types (Reyes-Palomeque et al., 2019) or structural variables (Andres-Mauricio et al., 2021) for AGBD modeling.

The texture metrics obtained from S1 and S2 serve as indicators of structural and spectral variability, respectively. These levels of variability are related to the homogeneity (low variability) or heterogeneity (high variability) of forest coverage, whose relationship with AGBD depends on the vegetation type and the texture metric used. In dense vegetation with a closed canopy, a positive relationship of AGBD with heterogeneity measures would be expected, given that greater structural heterogeneity is associated with a more complete occupation of the vertical profile of the vegetation and, therefore, with higher AGB values. However, in arid and semi-arid areas, zones with high AGB values tend to correspond to areas with a relatively more closed and structurally more homogeneous canopy compared to more open areas.

Regarding texture metrics, the two most important variables in the present study were NDVI_savg and B4_savg (Fig. 8), these being measures of homogeneity. In the first, a high positive correlation with AGBD was observed (0.48), suggesting that high AGBD values were estimated when NDVI values were homogeneous. On the other hand, the correlation of B4_savg with AGBD was observed at the same level as the former, but with a negative sign (−0.48), suggesting that high AGBD values occurred under heterogeneous conditions in B4. These results make sense when considering that NDVI is positively related to AGBD (Rodríguez-Veiga et al., 2017), while reflectance in the red channel (B4) decreases when AGBD is high because the photosynthetic pigments of plants absorb light in that region of the electromagnetic spectrum (Turner et al., 2003).

The third most important variable (NDVI_idm: Inverse Difference Moment) was negatively correlated with AGBD (−0.42; Fig. 8). Since

NDVI_idm corresponds to a homogeneity metric, this contradicts our observations for NDVI_savg. The mixed results regarding the correlation of texture metrics with AGBD could be due to the variation in structural and compositional conditions between the different vegetation types in this study. For example, as discussed above, homogeneity could be positively related to AGBD in xeric shrubland areas, where vegetation is more open and spatially dispersed, while heterogeneity could be positively related to AGBD in coniferous forests, with denser and more closed vegetation.

4.3. Comparison with other studies

The methodological approach used in this study significantly improved the accuracy of AGBD estimates compared to previous studies (Cartus et al., 2014; Rodríguez-Veiga et al., 2016), which obtained RMSE values that almost tripled those of the map of this study (Fig. 10c). These results support the third prediction made in this study, since AGBD estimates performed much better than those of previous studies and significantly reduced underestimation.

Several reasons can explain these improved results. First, this study used GEDI LiDAR data, which are distributed throughout the study area and, due to the two-stage modeling approach, allowed obtaining a very large sample size for model training and validation (38,949 1-ha cells). Since LiDAR pulses can penetrate the vegetation canopy, they are expected to underestimate AGBD to a lesser extent than optical imaging or SAR used in previous studies. Second, unlike previous studies, both the dataset used in the first stage for fitting the footprint model ($n = 53$) and the NFI data included a correction of AGBD for small individuals, shrubs, and cacti. Finally, the present study reduced the temporal discrepancy between the date of field data collection and the date of satellite image acquisition, which is one of the sources of error reported by Rodríguez-Veiga et al. (2016).

5. Conclusion

The GEDI project offers an unprecedented opportunity to generate estimates of forest attributes using LiDAR data. This study highlights the advantages of fitting new AGBD estimation models and shows the potential of generating field data with the same location of GEDI footprints to reduce georeferencing errors. This approach improved the accuracy of estimates from global models in the study area; however, given the high costs of making additional field measurements, the use of NFI data to increase the sample size and improve the models to predict footprint AGBD needs to be further explored. The AGBD values found in this study are much higher than those previously reported; we expect this will contribute to reassess the importance of vegetation in arid and semi-arid zones, as well as the need to densify and stratify NFI field samplings to produce more accurate and reliable estimates.

By associating GEDI AGBD estimates with different variables from satellite imagery, this study generated a wall-to-wall AGBD map with a 10 m spatial resolution, which significantly improved previous AGBD estimates for the study area. These maps can be useful for the design, implementation, and evaluation of REDD + strategies in the study area and in other arid and semi-arid areas.

The two-stage mapping approach employed in this study allowed increasing the sample size with LiDAR data and performing a wall-to-wall AGBD estimation with satellite imagery with a considerably large training and validation sample size, which, together with the corrections made to the field data, reduced the AGBD underestimation errors, mainly in arid areas. The results of the present study set the baseline for estimating the C stored in arid zones of Mexico, whose role in the mitigation of climate change may be seriously underestimated.

CRedit authorship contribution statement

Luis A. Hernández-Martínez: Writing – original draft,

Methodology, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Juan Manuel Dupuy-Rada:** Writing – review & editing, Writing – original draft, Supervision, Methodology. **Alfonso Medel-Narváez:** Writing – review & editing, Methodology. **Carlos Portillo-Quintero:** Writing – review & editing, Methodology. **José Luis Hernández-Stefanoni:** Writing – review & editing, Writing – original draft, Supervision, Methodology, Investigation, Conceptualization.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.srs.2025.100204>.

Data availability

Data will be made available on request.

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